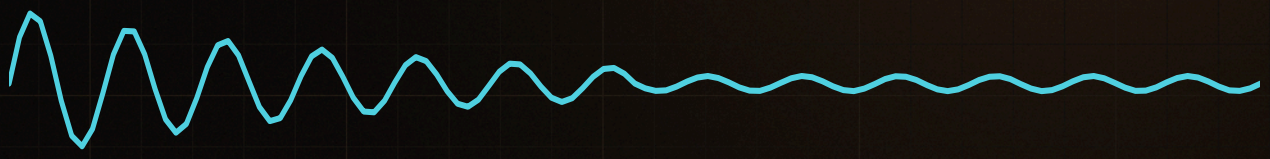
CHANNEL 02 · ACTIVE CH-02 · 868 MHz



CARRIER ACTIVE // 868 MHz · CYAN TRACE

CLASSIFIED — PROBLEM STATEMENT CHANNEL

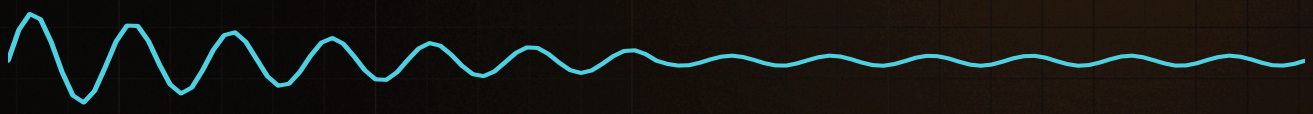
DEFENCE & NATIONAL SECURITY

PROBLEM STATEMENTS · ROUND 1

24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29–30 JANUARY 2027 · FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY, VASHI

DOMAIN SLOT 02 CH-02 · 868 MHz



CARRIER ACTIVE // 868 MHz · CYAN TRACE

DEFENCE & NATIONAL SECURITY

Ruggedized, mission-critical engineering that protects infrastructure, secures communication lines and assists emergency response. Resilient architectures for hostile, remote or constrained environments.

— AREAS TO EXPLORE

- Reconnaissance Robotics
- Encrypted Edge Comms
- Perimeter Telemetry
- Search & Rescue Tech
- Threat Detection

— WHO SHOULD EXPLORE

Robotics · Embedded Systems · RF Communications · Cyber-Physical Security & Rugged Electronics

> *"High stakes and zero tolerance for failure — engineer systems that stay online when everything else goes down."*

— IN THIS CHANNEL

PS-01	Portable Hybrid Solar-Battery Power Unit for Remote Field Operations
PS-02	Portable Communication Link Failure Detection & Alert System
PS-03	Portable Water Quality & Availability Monitoring for Remote Camps
PS-04	Drone-Based Inspection of Remote Infrastructure & Difficult Terrain
PS-05	Portable Emergency Lighting & Power Support System for Field Operations

PS-01 PORTABLE HYBRID SOLAR-BATTERY POWER UNIT FOR REMOTE FIELD OPERATIONS

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Remote posts, patrol camps and temporary shelters often have no grid supply and rely on diesel generators that are noisy, heavy to resupply and easy to detect. The challenge is to design a portable power unit that combines solar generation with battery storage to run essential loads such as radios, LED lighting and device chargers, and to size and control it for limited weight and harsh conditions.

B · CONTEXT

Fuel logistics to remote locations are costly and risky, and generator noise and heat can compromise a position. Solar-battery units offer silent, low-maintenance power but must be light enough to carry, simple to set up and robust against dust, rain and temperature extremes. Careful load budgeting and charge control determine whether such a unit actually lasts through cloudy days and nights.

C · CORE FEATURES

- Convert solar energy into stored battery energy using a charge controller (PWM or MPPT).
- Supply regulated outputs, e.g. 12 V DC, 5 V USB and, optionally, a small 230 V AC output.
- Protect against overcharge, deep discharge, overload, short circuit and reverse polarity.
- Indicate battery state of charge, solar input and load status clearly, even with gloved hands.
- Provide an optional secondary charging input (vehicle or hand crank).

D · DESIGN REQUIREMENTS

- Users: patrol teams, border outposts, disaster-relief and expedition parties.
- Portability: a justified mass/volume target (e.g. carried by one person), with quick set-up and pack-down.
- Environment: wide temperature range, dust and rain, tolerance of shock and vibration.
- Cost and upkeep: affordable, field-repairable with minimal tools, silent with a low thermal signature.

E · TECHNICAL FOCUS

LOAD BUDGET Daily energy demand (Wh) from a defined load list with duty cycles; size PV area, battery capacity and autonomy days from stated solar-hour assumptions (e.g. 4–6 peak sun hours).

POWER ELECTRONICS Charge controller, DC-DC converters and optional inverter; efficiency estimates for each stage.

BATTERY & PROTECTION Chemistry choice (e.g. LiFePO4 or other) with justified protection limits and temperature handling.

CONTROL & DISPLAY Microcontroller or controller IC, SoC estimation and a simple user display.

SIMULATION Sizing and charge-discharge simulation over several days in MATLAB/Simulink, Python or equivalent, using realistic or varied solar-input profiles.

OPTIONAL MPPT tracking, foldable panel design, vehicle charging, remote status link, thermal management.

F · EXPECTED OUTCOME

- Block diagram and a worked load-budget and sizing calculation.
- Simulation of solar input, battery state and load coverage over at least three days, including a low-sun day.
- Estimated mass, volume and cost of the proposed design.
- Protection scheme and failure-mode summary.
- Comparison with a diesel generator in weight, noise and logistics.

PS-02 PORTABLE COMMUNICATION LINK FAILURE DETECTION & ALERT SYSTEM

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Field teams depend on radio, cellular, satellite or Wi-Fi links to stay in contact. Link quality degrades with distance, terrain, weather, interference or equipment faults, and a link may fail without the users realising it. The challenge is to build a portable system that continuously monitors the health of a communication link, detects degradation or failure quickly and alerts the operator.

B · CONTEXT

A silent communication failure can leave a team unaware that messages are not getting through, creating risk when coordination is critical. Simply watching signal-strength bars is unreliable, because a strong signal can still carry no data. A portable monitor that combines several indicators — received signal strength, packet loss, delay and periodic heartbeat messages — can recognise problems early and provide an unmistakable local alert.

C · CORE FEATURES

- Send and receive periodic heartbeat messages between nodes and check that they arrive on time.
- Measure link-quality indicators such as signal strength (RSSI), packet loss and round-trip delay.
- Detect degradation and failure using thresholds with hysteresis to avoid flickering alerts.
- Alert the operator locally with a buzzer, LED or vibration, and show outage duration.
- Log link history and optionally indicate when a backup path is available.

D · DESIGN REQUIREMENTS

- Users: patrol teams, convoys, remote camps and emergency responders.
- Portability and endurance: small, light and low-power, running for days on a battery.
- Usability: alerts unmistakable in noise or darkness, usable by non-technical users.
- Compatibility and cost: low cost, usable with different link types, without interfering with the link.

E · TECHNICAL FOCUS

LINK MEASUREMENT RSSI, signal-to-noise ratio, packet-delivery ratio and latency from the radio module or heartbeat messages. Link technology is the participants' choice (e.g. LoRa, Wi-Fi, 2.4 GHz modules, cellular).

PROCESSING & LOGIC Microcontroller with a state machine (healthy, degraded, lost), moving-average filtering, timeouts and hysteresis; justify threshold values.

ALERT & INTERFACE Buzzer, LED indicators, vibration motor and small display.

POWER SUPPLY Battery with a stated runtime and low-power sleep strategy.

SIMULATION Link model with path loss (e.g. log-distance), shadowing, fading, interference and obstacles in Python or MATLAB; show how indicators behave as range grows or the link is blocked.

OPTIONAL Automatic switch to a backup link, GPS location stamp with failure events, mesh-network health map, long-term link statistics.

F · EXPECTED OUTCOME

- System architecture and a flowchart of the detection logic.
- Simulation of gradual degradation, sudden loss and intermittent interference, with system state over time.
- Detection time and false-alarm rate for the chosen thresholds.
- Power budget and expected battery life.
- Cost estimate and discussion of limitations.

PS-03 PORTABLE WATER QUALITY & AVAILABILITY MONITORING FOR REMOTE CAMPS

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Remote camps draw water from tanks, streams, wells or delivered supplies. Staff must know both how much water remains and whether it is safe to drink, with no laboratory, limited power and often no network. The challenge is to design a portable, rugged monitor that measures water quality and quantity at one or more sources and forecasts how many days of safe supply remain.

B · CONTEXT

Running short of water or drinking contaminated water can quickly disable a group, yet field monitoring is usually manual and sporadic. Conditions vary between sources and change with weather: a stream can turn turbid after rain, and tank levels fall faster than expected when headcount rises. A self-contained device that tracks level, usage and quality, and warns early, supports better rationing, treatment and resupply decisions.

C · CORE FEATURES

- Measure tank or reservoir level and estimate consumption rate.
- Forecast days of supply remaining at current usage and flag a low-supply condition.
- Measure key quality indicators (turbidity, pH, TDS/conductivity, temperature), optional chlorine/ORP for treated water.
- Compare multiple sources and classify each as suitable, treat before use, or unsafe.
- Operate offline with local display and log data for later retrieval.

D · DESIGN REQUIREMENTS

- Users: field camps, outposts, relief shelters and expedition parties.
- Ruggedness: portable, dust- and splash-resistant, tolerant of temperature swings.
- Power and connectivity: low-power on battery or small solar, working without a network.
- Practical value: simple go/no-go indication for non-specialists, with clear statements of the device's limits (not a certified lab test).

E · TECHNICAL FOCUS

LEVEL SENSING Ultrasonic, pressure or float sensing suited to the tank type; discuss accuracy and calibration for different tank shapes.

QUALITY SENSING Turbidity, pH, conductivity/TDS and temperature probes with temperature compensation, calibration interval and probe-care considerations.

ANALYTICS Consumption-rate estimation with filtering (to ignore refilling and noise), days-remaining forecast and rule-based quality classification.

PROCESSING, COMMS & POWER Low-power microcontroller; local display; optional LoRa or GSM status reports; battery and small solar panel.

SIMULATION Simulated multi-day scenario in Python or MATLAB with varying headcount, rainfall events, refill events and source-quality changes.

OPTIONAL Water-treatment dosing guidance, flow measurement at sources, camp-wide dashboard, ML forecasting of demand.

F · EXPECTED OUTCOME

- System architecture, with sensor choices justified for field use.
- Simulation showing level, consumption estimate and days-remaining forecast with its error.
- Quality-classification logic with example outputs for good, turbid and contaminated sources.
- Power budget and expected battery life.
- Cost estimate, calibration plan and limitations.

PS-04 DRONE-BASED INSPECTION OF REMOTE INFRASTRUCTURE & DIFFICULT TERRAIN

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Remote infrastructure such as perimeter fencing, bridges, communication towers, power lines, pipelines and mountain roads is spread over long distances and rough terrain that is slow, tiring or dangerous to reach. The challenge is to design a drone-based survey system that can cover long routes over uneven ground within its battery limits, keep contact with the operator and return useful inspection data.

B · CONTEXT

Ground patrols take days and expose personnel to terrain and weather hazards, while satellite images are infrequent and coarse. Drones can survey long linear assets quickly, but real routes involve elevation changes, wind, limited battery endurance, communication range limits and no landing options. Careful mission planning, an energy and link budget, failsafe behaviour and efficient data handling decide whether such a system is practical and safe.

C · CORE FEATURES

- Plan routes over terrain with elevation data, waypoints, altitude and camera coverage.
- Estimate flight energy and range under payload, wind and climb conditions, and plan battery swaps or recharging stops.
- Maintain command and telemetry links, with defined behaviour on link loss, low battery or poor weather.
- Capture visual or thermal images, tag them with location and flag anomalies such as damage, intrusion or erosion.
- Produce a concise inspection report and a map of findings.

D · DESIGN REQUIREMENTS

- Users: infrastructure operators, security and border-management agencies, disaster-management teams.
- Operational value: longer coverage per sortie, fewer physical patrols, faster response after events.
- Compliance and safety: adherence to civil drone regulations and restricted-airspace rules; designs for authorised use only.
- Cost and logistics: affordable platform, simple field set-up, transport and maintenance.

E · TECHNICAL FOCUS

PLATFORM & ENERGY Multirotor or fixed-wing choice justified by range, hover needs and payload; energy model including climb, wind and reserve margin.

NAVIGATION & LINK GNSS, IMU, barometer, geofence and return-to-home logic; radio link budget (frequency, power, antennas, terrain shadowing) and relay options.

PAYLOAD & DATA Visual and optional thermal camera, on-board storage and selective data transfer; simple image-based anomaly detection welcome.

MISSION PLANNING Terrain-following route generation and coverage calculation from digital elevation data or synthetic terrain.

SIMULATION Flight energy and coverage simulation in MATLAB, Python or a UAV simulator (e.g. PX4 SITL with Gazebo), including link-loss and low-battery scenarios.

OPTIONAL Autonomous battery-swap station, mesh relay drones, change detection between surveys, 3D terrain reconstruction.

F · EXPECTED OUTCOME

- System architecture covering aircraft, link, ground station and data flow.
- Mission plan for a representative route (e.g. 20–30 km of linear infrastructure) with battery and range analysis.
- Link-budget analysis and a simulation of at least one failure scenario.
- Demonstration of anomaly flagging on sample images, with performance measures.
- Cost estimate and comparison with a ground-patrol approach.

PS-05 PORTABLE EMERGENCY LIGHTING & POWER SUPPORT SYSTEM FOR FIELD OPERATIONS

Recommended for Third & Final Year students

— A · PROBLEM DEFINITION

When primary power or lighting fails in a field camp, shelter or command post, teams need immediate, dependable light and electrical power for communication and essential equipment. The challenge is to design a portable emergency system that detects failure and activates autonomously, manages several energy sources and loads intelligently, and keeps the most critical functions alive for a defined mission time in harsh environments.

— B · CONTEXT

Field emergency kits typically offer a battery-powered lamp and a small charger with no coordination between them, so a light left on can drain the energy needed for radios or medical equipment. Operations may run for many hours in heat, cold, dust and rain, with energy arriving from solar, a vehicle, a hand crank or a generator. Intelligent power sharing, adaptive lighting and fault tolerance become decisive.

— C · CORE FEATURES

- Detect primary-supply or lighting failure and switch on emergency light and power within a defined time.
- Combine multiple energy inputs (battery, solar, vehicle, hand crank, generator) with an arbitration scheme.
- Prioritise loads (communication, medical, lighting, charging) and degrade gracefully when energy is short.
- Provide adaptive lighting modes (high-output, low-glare, low-visibility) with ambient-light and motion-based dimming.
- Estimate remaining mission time and report battery health, faults and thermal state.

— D · DESIGN REQUIREMENTS

- Users: field camps, command posts, rescue teams and disaster-response units.
- Ruggedness: high ingress protection (e.g. IP65+), shock and drop tolerance, justified operating temperature range.
- Mission assurance: redundancy for critical functions, short-circuit and overload protection, fast field repair or module swap.
- Logistics: limited weight, modular configuration and reasonable cost for fielding in numbers.

— E · TECHNICAL FOCUS

ENERGY ARCHITECTURE Source and load topology, DC bus voltage choice, converters and protection; discuss conversion losses and redundancy.

CONTROL & ARBITRATION Microcontroller energy manager implementing source priority, load priority, shedding order and mode logic; define behaviour when sources vary unpredictably.

ESTIMATION & HEALTH SoC and remaining-time estimation, battery temperature handling at low and high temperatures, and fault diagnostics.

LIGHTING CONTROL LED driver with dimming and colour/low-signature modes, sensors for ambient light or occupancy.

SIMULATION 12–48 hour mission simulation in Simulink, Python or equivalent with changing load demand and energy-source availability, comparing the manager with a fixed-priority baseline.

OPTIONAL Wireless status reporting, vehicle-integrated charging, predictive health management, modular stackable battery packs.

— F · EXPECTED OUTCOME

- System architecture, energy-flow diagram and control-logic description.
- Mission simulation showing source use, load shedding and battery state over 12–48 hours.
- Quantified benefit of the manager over the baseline (e.g. extra hours of critical-load operation).
- Failure-mode analysis covering source loss, short circuit and battery fault, with the system's response.
- Mass, volume and cost estimate, and a discussion of environmental robustness.